Mixed Boundary-Value Problems: Modelling and Method Design

Dirichlet / Neumann / Robin Poisson Problem — Boundary Closures, Solver Strategy, and Convergence-Study Design

Direction 2 — Mixed boundary data **Module** Modelling (this document) / Implementation (follow-up) **Date** October 2026

Deliverable mapping: `problem_definition_sheet` Part 1 + Part 2 (FDM stencil) Scope: core problem + Depths 1–4 + L-shaped innovation Notation aligned with the course lecture notes

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0 · Executive Summary

The model in one sentence: solve the Poisson equation $-\Delta u = f$ on the unit square with a **Dirichlet–Neumann (or Dirichlet–Robin) mixed boundary**, discretised by the five-point finite-difference method; the open right edge is where the real numerical work lies — **the boundary closure itself is part of the method**.

Within the team division of labour, this module (modelling + programming) delivers: a precise definition of the continuous problem, coefficient-level derivations of the discretisations and boundary closures, the algebraic-solver strategy, and the complete experiment design for the four depth extensions and the L-shaped innovation. This document is the modelling deliverable and simultaneously feeds `problem_definition_sheet.tex` Part 1 (continuous problem, sign conventions, method choice) and Part 2 (discrete operator: stencil).

Five experiments (details in §8):

ID	Experiment	Setup	Key predictions (written before running)
E1	Two-closure comparison (core)	{Neumann, Robin} × {one-sided, ghost-point}, $N = 8, 16, 32, 64$	one-sided: boundary L_∞ order $\to 1$; ghost: global $L_h^2/L_\infty \to 2$
E2	Direct vs iterative (Depth 2)	same mesh family, matched scaled-residual accuracy target	SOR(ω^*) counts $O(h^{-1})$, Jacobi $O(h^{-2})$; the direct solver is expected to win throughout the tested range — report the absence of a crossover explicitly
E3	FDM vs FEM (Depth 3)	rectangular pure-Dirichlet Poisson benchmark, same mesh, same h, same tolerance	both L_h^2 second order; DOFs strictly identical $(N-1)^2$; identical nonzero patterns ($h^2 A_{\mathrm{FDM}} = K_{P1}$)
E4	Smooth vs singular (Depth 4)	two analytic solutions with $f = 0$ on the L-shaped domain: $e^x \cos y$ and $r^{2/3} \sin \frac{2\varphi}{3}$	smooth → order 2; singular: $L_\infty \to \frac{2}{3}$, $L_h^2 \to \frac{4}{3}$ (measured FDM results)
E5	Graded meshes (innovation, advanced)	red-refinement pyramid as the main route, radial grading μ-sweep as theory validation	uniform energy error $O(N_{\mathrm{dof}}^{-1/3})$; grading recovers $O(N_{\mathrm{dof}}^{-1/2})$, threshold $\mu > 1/\lambda = 3/2$

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1 · Background: Why Mixed Boundary Data Is the Real Problem

1.1 Physical Meaning of the Three Boundary-Condition Types

The standard mathematical model of steady diffusion (heat conduction, electrostatics, porous-media pressure) is the Poisson equation $-\Delta u = f$. Real devices rarely prescribe the same type of condition on the whole boundary — **mixed boundary data is the engineering norm**:

Table 1.1 The three boundary-condition types: mathematical form, physical prototype, and their role in this problem

Type	Mathematical form	Physical prototype	Position in this problem
Dirichlet	$u = g$ on Γ_D	isothermal reservoir, fixed potential, known concentration	left edge $x = 0$, lower edge $y = 0$, upper edge $y = 1$ (**including all four corners**)
Neumann	$\partial_n u = q$ on Γ_B	prescribed heat flux, insulated wall ($q = 0$), injected current	open right edge $x = 1$, variant A
Robin	$\partial_n u + \beta u = r,\ \beta > 0$	convective heat exchange (Newton cooling), linearised radiation, contact resistance	open right edge $x = 1$, variant B, $\beta = 2$

Two application scenarios deserve emphasis in the report (they motivate the "phenomenon" narrative):

- **Heat exchangers**: one wall segment held at a phase-change temperature (Dirichlet), another insulated or flux-loaded (Neumann), and a segment in convective exchange with the fluid, $q = \alpha(T - T_\infty)$ (Robin, i.e. the $\partial_n u + \beta u = r$ form).
- **Pressure-Poisson equations** (CFD projection methods): once the momentum equation yields the velocity, the pressure is recovered from $-\Delta p = \nabla \cdot \mathbf{f}$, and its normal-gradient data $q = \partial_n p$ **comes from the momentum equation projected onto the boundary** — it is not a gauge choice that may be set to zero. Setting $p = 0$ along an entire edge is a **physical Dirichlet condition**, not a gauge. This is the core of the "phenomenon" narrative and a misunderstanding the problem pack explicitly asks the report to dispel.

1.2 Well-Posedness and the Variational Framework

Take $\Omega = (0,1)^2$ and $f \in L^2(\Omega)$. Following the two-step derivation chain from the FEM lecture notes — multiply by a test function v, integrate, and use the general identity (integration by parts transfers one derivative from u to v, so that piecewise-linear functions with piecewise-constant first derivatives become usable):

$$\int_\Omega (-\Delta u)\, v \, dx \;=\; \int_\Omega \nabla u \cdot \nabla v \, dx \;-\; \int_{\partial\Omega} \partial_n u \; v \, ds.$$

The boundary term is treated segment by segment: on the **Dirichlet part** we take $v = 0$, so the term vanishes (v is an "admissible perturbation direction that keeps the boundary values"); on the **Neumann part**, $\partial_n u = q$ enters the right-hand side as data (the lecture load formula $F_i = \int_\Omega f\varphi_i\,dx + \int_{\Gamma_N} q\varphi_i\,ds$); on the **Robin part**, $\partial_n u = r - \beta u$, and the term $-\beta u\, v$ moves to the left-hand side as the stiffness contribution $\int_{\Gamma_B} \beta uv\,ds$. Writing $\partial_n u = r - \beta u$ uniformly (variant A is $\beta = 0$, $r \equiv q$), the weak form reads: find $u \in H^1(\Omega)$ with $u|_{\Gamma_D} = g$ such that

$$\int_\Omega \nabla u \cdot \nabla v \, dx + \int_{\Gamma_B} \beta uv \, ds \;=\; \int_\Omega fv \, dx + \int_{\Gamma_B} rv \, ds \quad \forall\, v \in H^1_0(\Omega;\Gamma_D),$$

where $H^1_0(\Omega;\Gamma_D) = \{v \in H^1 : v|_{\Gamma_D} = 0\}$. Neumann/Robin data enter as **natural boundary conditions** — they never appear in the trial space (this is precisely why FEM handles derivative conditions "naturally" while FDM needs a dedicated closure). By Lax–Milgram (the bilinear form satisfies $a(u,v) = \int_\Omega \nabla u\cdot\nabla v + \int_{\Gamma_B}\beta uv \geq \|\nabla v\|^2_{L^2}$, together with the Poincaré inequality enabled by $\Gamma_D \neq \varnothing$; inhomogeneous g is handled via the trace theorem):

- Γ_D **nonempty (our main problem)**: a **unique solution** exists; no compatibility condition is required.
- **Pure Neumann** (optional depth): $-\Delta u = f$ with $\partial_n u = q$ is solvable only under the **compatibility condition** $\int_\Omega f\,dx + \int_{\partial\Omega} q\,ds = 0$, and the solution is **determined only up to a constant** (nullspace span$\{1\}$); the mean or one nodal value must be fixed afterwards (a gauge).
- **Robin with** $\beta > 0$: contributes $\int_{\Gamma_B}\beta uv\,ds \geq 0$ to the bilinear form and **removes the constant nullspace even when** Γ_D **is empty**, eliminating the compatibility condition — a piece of "extra mathematical value" of Robin data worth pointing out in the report.

1.3 The Numerical Challenge: The Boundary Closure *is* Part of the Method

This is the essential difference from pure Dirichlet problems and the methodological through-line of the whole study. For Dirichlet data, "move the known values to the right-hand side" is all there is; Neumann/Robin data constrain **derivatives**, and several inequivalent algebraic choices exist (first-order one-sided quotient? second-order centred quotient? a ghost point?) — each choice **changes the row structure of the linear system**:

- **Row-structure change**: the one-sided closure **replaces** the boundary node's PDE row by a derivative row of size h^{-1}; the ghost closure **retains** the boundary PDE row (size h^{-2}) and only eliminates the auxiliary exterior value. This is not merely "first order vs second order formula" — it is a difference in **how the equation system is constituted**. The problem pack explicitly requires this difference of formulation to be stated in the report, and forbids attributing every observed difference to the order of the derivative formula.
- **Loss of symmetry**: in either closure the right-edge rows scale differently from the interior PDE rows (ghost elimination doubles the inward coupling to -2), and the assembled matrix is **nonsymmetric**. "The continuous Laplacian is self-adjoint" is not a valid reason to use conjugate gradients — a methodological error the pack names explicitly.
- **Solvability structure**: the Neumann edge carries the latent threat of a zero-eigenvalue direction (defused here by the nonempty Γ_D), but understanding the pure-Neumann compatibility/nullspace structure is the necessary background for explaining why this problem does not need it.

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MODELLING NARRATIVE

Three-part background story: **(i)** boundary data are naturally mixed in engineering (heat exchangers / pressure Poisson); **(ii)** mathematically, well-posedness is guaranteed by $\Gamma_D \neq \varnothing$, and Robin has additional value; **(iii)** numerically, the closure choice itself constitutes a method difference — leading to the two technical through-lines of this study: the two-closure comparison and the solver strategy for nonsymmetric matrices.

2 · Continuous Model Specification

The specification below follows the conventions of `problem_definition_sheet.tex` Part 1; all symbols are used uniformly across the project.

2.1 Problem Statement and Sign Conventions

$$-\Delta u = f \quad \text{in } \Omega = (0,1)^2, \qquad u = g \text{ on } \Gamma_D, \qquad \begin{cases} \partial_n u = q & \text{variant A (Neumann)} \\ \partial_n u + \beta u = r,\ \beta = 2 & \text{variant B (Robin)} \end{cases} \text{ on } \Gamma_B$$

Boundary segmentation and ownership (**corner ownership is part of the specification, not an implementation detail**):

Table 2.1 Boundary segments and data

Segment	Geometry	Condition type	Data (manufactured from u^*)
Γ_D	the three edges $x=0$, $y=0$, $y=1$, **including all four corners**	Dirichlet	$g = u^*\|_{\Gamma_D}$
Γ_B	the open right edge $x=1$, corners excluded)	Neumann or Robin	$q(y) = e^{1+y}$; $r(y) = 3e^{1+y}$

Sign conventions (uniform throughout; the report must state them explicitly):

- the equation is written $-\Delta u = f$ (minus sign outside), so the assembled matrix has positive diagonal entries and is a weakly diagonally dominant M-matrix (see §3.6);
- $\partial_n u = \nabla u \cdot \mathbf{n}$ with $\mathbf{n}$ the **unit outward normal**. On the right edge $x = 1$, $\mathbf{n} = (1, 0)$, so $\partial_n u = u_x(1, y)$ — **the direction is positive** x. The datum $q = e^{1+y} > 0$ means u increases along the outward normal, which serves as a sign self-check;
- the corners $(0,0), (1,0), (0,1), (1,1)$ are all **owned by Dirichlet**: the right-edge corners $(1,0)$ and $(1,1)$ are not unknowns of a Neumann/Robin row. This convention removes any ambiguity about which row is assembled at a corner.

2.2 Manufactured Solution and Data Consistency

$$u^*(x,y) = e^{x+y} \implies f = -\Delta u^* = -2e^{x+y}, \qquad g = u^*|_{\Gamma_D}, \qquad q(y) = u^*_x(1,y) = e^{1+y}, \qquad r(y) = u^*_x(1,y) + 2\,u^*(1,y) = 3e^{1+y}.$$

Method of Manufactured Solutions (MMS): pick a smooth u^*, compute f and the boundary data backwards, so the numerical solution has a pointwise-computable reference. **Independent verification** of data consistency (mandatory in the report): $r = q + \beta u^*(1,y) = e^{1+y} + 2e^{1+y} = 3e^{1+y}$ √. The choice $u^* = e^{x+y}$ is not polynomial, avoiding the zero-error trap of low-degree polynomials being reproduced exactly by the five-point stencil (the course sanity check explicitly warns against $\pm(x^2 - y^2)$-type solutions).

2.3 Solvability Summary

- Variants A and B both have a nonempty Γ_D: **unique solution**, no compatibility condition, no numerical gauge issue;
- Optional depth (pure Neumann): after turning Γ_D into Neumann data, two effects must be handled discretely — **(a)** the discrete compatibility defect $\sum_{\text{nodes}} h^2 f + \sum_{\text{right edge}} h\, q_b \neq 0$ (a quadrature mismatch between the rectangle rule and the boundary line integral), requiring a projection correction of the right-hand side; **(b)** the constant vector $\mathbf{1}$ lies in the nullspace, so the mean or one nodal value must be fixed after the solve (gauge). The lecture-note argument for the nullspace: every five-point row is invariant under a global shift — adding a constant C to all five nodes changes the left-hand side by $\frac{C}{h^2}(4 - 1 - 1 - 1 - 1) = 0$; the Neumann difference row $U_B - U_P = hq$ is shift-invariant as well; hence all row sums vanish, $A\mathbf{1} = \mathbf{0}$, $\det A = 0$ (singular), and the solution is at most determined up to $U_0 + C\mathbf{1}$. Fixing the mean or one nodal value restores uniqueness. Experiment design: §8-E6.

3 · Discrete Model (FDM): Grid, Five-Point Stencil, and Two Boundary Closures

3.1 Grid, Unknowns, and Indexing

Nodal grid $x_i = ih$, $y_j = jh$, $h = 1/N$, $i, j = 0, \ldots, N$. The set of unknowns:

- **PDE-row unknowns** $U_{i,j}$ with $1 \le i \le N-1$, $1 \le j \le N-1$, $(N-1)^2$ in total (five-point PDE rows are assembled at these nodes); the **right-edge nodes** $i = N$ **($1 \le j \le N-1$) are also unknowns** (Neumann/Robin values are not eliminated; closure rows are assembled there). The total number of unknowns is therefore $n = (N-1)^2 + (N-1) = N(N-1)$. Uniform notation: $\{U_{i,j} : 1 \le i \le N,\ 1 \le j \le N-1\}$, $N(N-1)$ in total.
- **Known values**: the columns/rows $i = 0$, $j = 0$, $j = N$ (Dirichlet, imposed and eliminated), plus the two right-edge corners $(1,0)$ and $(1,1)$.
- **Lexicographic index** $k = (j-1) \cdot N + (i-1)$ (row-major; each row j contains N consecutive unknowns $i = 1, \ldots, N$), so the matrix is banded with half-bandwidth $\approx N$. The index map $(i,j) \mapsto k$ is implemented as an explicit function, which simplifies assembly and error back-substitution.

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3.2 Interior Five-Point Stencil

For $-\Delta u = f$ at node $P = (x_i, y_j)$ use centred differences; Taylor expansion gives

$$\frac{u_E - 2u_P + u_W}{h^2} = u_{xx}\big|_P + \frac{h^2}{12}u_{xxxx}\big|_P + O(h^4), \qquad \text{summing both directions:} \quad -\Delta_h u_P = f_P - \frac{h^2}{12}(u_{xxxx} + u_{yyyy})\big|_P + O(h^4).$$

The five-point row ($1 \le i \le N$, $1 \le j \le N-1$; the eastern neighbour at $i = N-1$ is the right-edge unknown, and $i = N$ is treated in the next subsection):

$$\boxed{\frac{1}{h^2}\Big(4U_{i,j} - U_{i-1,j} - U_{i+1,j} - U_{i,j-1} - U_{i,j+1}\Big) = f_{i,j}} \qquad \text{(interior PDE row)}$$

Local truncation error $\tau_P = -\frac{h^2}{12}(u_{xxxx} + u_{yyyy})|_P + O(h^4)$: for $u^* = e^{x+y}$ we have $u_{xxxx} + u_{yyyy} = 2e^{x+y}$, hence $\tau_P = -\frac{h^2}{6}e^{x_i+y_j} + O(h^4)$ — exactly computable, and usable as a magnitude cross-check for the E1 error predictions.

3.3 Dirichlet Elimination and Corner Ownership

All Dirichlet node values are substituted into neighbouring equations; known contributions enter the right-hand side multiplied by their coefficient $-1/h^2$: e.g. a row with $j = 1$ contains $U_{i,0} = g_{i,0}$, so the RHS gains $g_{i,0}/h^2$. **Assembly-order rule** (written into code and report): first draft "PDE rows" at all nodes, then **replace or rewrite the right-edge rows as whole rows** according to the chosen closure, and finally eliminate the Dirichlet knowns. **Corner protection**: right-edge rows are assembled only for $1 \le j \le N-1$; the right-edge nodes with $j = 0$ and $j = N$ carry no unknowns, so "a corner overwritten by a Neumann/Robin row" is structurally impossible.

3.4 Closure A: First-Order One-Sided Rows

Idea: discretise the boundary condition directly with a **first-order backward (inward) difference** and use that derivative row to **replace** the boundary node's PDE row.

$$\text{Neumann:} \quad \frac{U_{N,j} - U_{N-1,j}}{h} = q_j \quad \Longrightarrow \quad \frac{1}{h}U_{N,j} - \frac{1}{h}U_{N-1,j} = q_j$$

$$\text{Robin:} \quad \frac{U_{N,j} - U_{N-1,j}}{h} + \beta U_{N,j} = r_j \quad \Longrightarrow \quad \Big(\frac{1}{h} + \beta\Big)U_{N,j} - \frac{1}{h}U_{N-1,j} = r_j$$

Local truncation of this row (substituting u^* into the left side minus the data): the backward-difference expansion gives the residual $\frac{u(1)-u(1-h)}{h} - u_x(1) = -\frac{h}{2}u_{xx}(1) + O(h^2)$, i.e. $O(h)$ **consistency**. For u^* this can be computed exactly:

$$\frac{u^*(1, y_j) - u^*(1-h, y_j)}{h} - q_j = e^{1+y_j}\Big(\frac{1 - e^{-h}}{h} - 1\Big) = -e^{1+y_j}\,\frac{h}{2} + O(h^2) \;< 0.$$

Note that this row **no longer contains** f — the PDE at the boundary node is completely replaced by a derivative relation, and $f_{N,j}$ never appears in the equation. This is the concrete embodiment of the "how the equation system is constituted" difference of §1.3.

> **LECTURE-NOTE ALIGNMENT**
>
> The lecture notes on one-dimensional differences teach single-sided Neumann via the "**bulk-difference substitution**" version: "the directional derivative times the step size gives a difference" $U_{N,j} = U_{N-1,j} + hq_j$, which is substituted directly into the five-point equation at the **adjacent interior node** (x_{N-1}, y_j), giving $3U_{N-1,j} - U_{N-2,j} - U_{N-1,j-1} - U_{N-1,j+1} = h^2 f_{N-1,j} + hq_j$ (diagonal $3/h^2$, right-hand side $+q_j/h$) — the right-edge node **carries no unknown** and is recovered afterwards from the difference relation. Both assemblies **yield exactly the same solution**: block-eliminating $U_{N,j}$ from this project's closure-A derivative row $U_{N,j} - U_{N-1,j} = hq_j$ (after scaling by h) produces precisely the lecture substitution row. The differences lie at the system level: ① **unknown set**: project convention $n = N(N-1)$ (right edge as unknowns) vs lecture version $(N-1)^2$ (boundary value eliminated); ② **matrix symmetry**: the lecture version (with all other edges Dirichlet) is actually **symmetric**, while the project convention is nonsymmetric — the pack's requirement to report $\|A - A^\top\|/\|A\|$ makes sense only under the project's assembly convention; ③ **row scaling**: the project convention keeps h^{-1}-scaled rows (the reason the pack demands scaled residuals), while the lecture version absorbs the boundary effect into interior rows.
>
> **Why the project convention is used**: the problem pack defines the right-edge unknowns $U_{N,j}$, $1 \le j \le N-1$, and stipulates "the derivative row replaces the PDE row"; the E1 quantities "right-edge error separation" and $\|A - A^\top\|/\|A\|$ are evaluable only when the right-edge nodes are unknowns. Explaining this block-elimination equivalence in the report both connects to the lecture and justifies the choice of assembly convention (the same physical boundary condition under different assembly protocols — a living example of the pack's "the comparison is deeper than the formula order").

3.5 Closure B: Second-Order Ghost-Point Rows

Idea: introduce a **conceptual exterior value** $U_{N+1,j}$ at $(1+h, y_j)$ (not an unknown), discretise the boundary condition with a **centred difference** to eliminate it, and **retain** the five-point PDE row at the boundary node.

Step 1 (boundary condition):

$$\frac{U_{N+1,j} - U_{N-1,j}}{2h} = q_j \implies U_{N+1,j} = U_{N-1,j} + 2h\,q_j \qquad \text{(Neumann)}$$

$$\frac{U_{N+1,j} - U_{N-1,j}}{2h} + \beta U_{N,j} = r_j \implies U_{N+1,j} = U_{N-1,j} + 2h\,(r_j - \beta U_{N,j}) \qquad \text{(Robin)}$$

Step 2 (substitution into the PDE row): impose the five-point equation at the boundary node $(1, y_j)$ and substitute the ghost value:

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$$\text{Neumann:}\quad \frac{4U_{N,j}-U_{N+1,j}-U_{N-1,j}-U_{N,j-1}-U_{N,j+1}}{h^2}=f_{N,j}$$

$$\Longrightarrow\ \boxed{4U_{N,j}-2U_{N-1,j}-U_{N,j-1}-U_{N,j+1}\ =\ h^2 f_{N,j}+2h\,q_j}\quad\text{(after multiplying by } h^2\text{)}$$

$$\text{Robin:}\quad 4U_{N,j}-\big[U_{N-1,j}+2h(r_j-\beta U_{N,j})\big]-U_{N-1,j}-U_{N,j-1}-U_{N,j+1}=h^2 f_{N,j}$$

$$\Longrightarrow\ \boxed{(4+2h\beta)\,U_{N,j}-2U_{N-1,j}-U_{N,j-1}-U_{N,j+1}\ =\ h^2 f_{N,j}+2h\,r_j}$$

Checks and observations:

- the inward coupling doubles to -2 (ghost elimination presses both the eastern and western contributions onto the western neighbour) — the nonsymmetry source named verbatim in the problem pack: "ghost elimination doubles the inward coupling";
- the Robin β term **strengthens** the main diagonal ($+2h\beta$) and gives the row sum $2h\beta>0$ — strictly diagonally dominant, isomorphic to the continuous mechanism "Robin removes the nullspace";
- the row keeps $f_{N,j}$ (its PDE-row identity is unchanged), and its scaling h^{-2} matches the interior rows — the root cause of "residuals are not comparable across meshes" against closure A's h^{-1} rows (see §3.7);
- residual on substituting u^*: the centred quotient $\frac{u(1+h)-u(1-h)}{2h}-u_x(1)=\frac{h^2}{6}u_{xxx}+O(h^4)$, i.e. for u^*, $e^{1+y_j}\big(\frac{\sinh h}{h}-1\big)=e^{1+y_j}\frac{h^2}{6}+O(h^4)-O(h^2)$ **consistency** √.

> **LECTURE-NOTE ALIGNMENT**
>
> The **order of operations** follows the lecture notes: "complete the stencil, express the ghost value from the boundary condition, then substitute back into the PDE". The lecture explanation of the second order: the symmetric left-right centring makes the **even Taylor terms cancel**, $\frac{u(x_B+h)-u(x_B-h)}{2h}=u'(x_B)+\frac{h^2}{6}u'''(x_B)+O(h^4)$, so the local truncation is $O(h^2)$, while the one-sided difference carries an $O(h)$ term $-\frac{h}{2}u''(x_B)$. Two lecture warnings enter the report: **(i)** "the local equation residual and the global solution error are different quantities" — the row-defect order and the global convergence order are reported separately, which is exactly what the §3.8 interior/boundary separation is designed for; **(ii)** the notes point out that after ghost elimination the boundary PDE row "may be only $O(h)$ while the global solution still converges second order" — under the present smooth solution the Appendix-A substitution check gives $O(h^2)$ components, and the **complete-row defect is** $O(h)$ (the $(2/h)$ amplification of the $O(h^2)$ boundary-condition residual produces the $\frac{h}{3}u'''$ term, report Eq (12)); E1 measures which face shows up in the data; **(iii)** ghost points also serve Dirichlet (linear interpolation $U_G=2g-U_P$ when the boundary lies between two nodes), Robin, and periodic conditions — the present Robin ghost row is a direct application of "the combination of normal derivative and function value fixes the ghost value".

3.6 Fundamental Contrast and Matrix Properties

Table 3.1 Contrast of the two closures (the pack explicitly demands this formulation difference in the report)

Aspect	Closure A: one-sided	Closure B: ghost-point
Equation identity of the boundary node	derivative row **replaces** the PDE row (f absent there)	PDE row **retained**; the ghost value eliminated via the BC
Local consistency order	$O(h)$ (residual $-\frac{h}{2}u_{xx}$)	$O(h^2)$ for the BC relation (residual $\frac{h^2}{6}u_{xxx}$); complete row $O(h)$ via the $(2/h)$ amplification
Row scaling	h^{-1} (**different scale** from interior h^{-2})	h^{-2} (matches interior rows)
Inward coupling	$-1/h$ (single neighbour)	$-2/h^2$ (doubled → visible nonsymmetry)
Robin diagonal correction	row $\times h$: diagonal $1+h\beta$, row sum $h\beta>0$	diagonal $4+2h\beta$, row sum $2h\beta>0$
Nonsymmetry	small row norm → weak in relative Frobenius measure	mismatch $-2/h^2$ vs $-1/h^2$ → more visible
Predicted global behaviour	boundary L_∞ order 1; global order 1 (to be confirmed by E1)	smooth data: global L_h^2, L_∞ order 2

Matrix-property checklist (verified item by item after assembly; constitutes the matrix argument of Part 2):

- interior rows: diagonal $+4/h^2$, off-diagonal $-1/h^2$, row sum 0 — weakly diagonally dominant, a Z-matrix;
- closure A Neumann row: diagonal $1/h$, off-diagonal $-1/h$, row sum 0 — weakly dominant; Robin row: row sum $h\beta>0$ — strictly dominant;
- closure B Neumann row: $4/h^2, -2/h^2, -1/h^2, -1/h^2$, row sum 0 — weakly dominant; Robin row: row sum $2h\beta/h^2>0$ — strictly dominant;
- all off-diagonal entries ≤ 0 and an irreducibly connected grid ⇒ all four assembled matrices are **irreducibly weakly diagonally dominant M-matrices**: nonsingular, nonnegative inverse, discrete maximum principle (the numerical solution stays between the boundary extrema — an independent E1 sanity check).

3.7 System Specification and Scaled Residuals

- System size $n=N(N-1)$ (PDE-row nodes + right-edge unknowns), at most 5 nonzeros per row, $\approx 5n$ in total, banded (half-bandwidth $\approx N$).
- **Nonsymmetry measure**: $\delta_A=\|A-A^\top\|_F/\|A\|_F$, tabulated against N (prediction: dominated by the $(-2/h^2)$ mismatch of the $O(N)$ boundary rows, δ_A decays under refinement but never vanishes — decay does not license "treat it as symmetric").
- **Scaled residual** (explicitly required by the pack): PDE rows scale like h^{-2}, closure-A rows like h^{-1}; scaling is mandatory before comparing residuals across meshes. With the row-scaling diagonal $D_s=\mathrm{diag}(s_k)$, s_k the row norm (or row absolute sum), report

$$\rho_{\text{scal}}=\frac{\|D_s^{-1}(b-Ax)\|_\infty}{\|D_s^{-1}b\|_\infty},$$

with **separate reports for the PDE-row and boundary-row groups**.

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- **Separation of algebraic and discretisation errors** (standing protocol): on a representative mesh, re-solve at 2–3 tighter tolerances and confirm the reported solution error stops changing; for the direct solver report the backward residual $\|b - Ax\|$ and a condition-number estimate $\kappa(A)$ (prediction: $\kappa \sim O(h^{-2})$).

3.8 Error Metrics and Convergence Protocol

Table 3.2 Unified convergence protocol (used by E1–E5)

Item	Definition / rule
Discrete L_h^2 norm	$\|e\|_{L_h^2} = \left(h^2 \sum_{\text{unknown nodes}} (U - u^*)^2\right)^{1/2}$, weight h^2 (nodal quadrature); the report **states weights and node set explicitly**
Nodal L_∞ norm	$\|e\|_\infty = \max_{\text{nodes}} \|U - u^*\|$ over all unknown nodes
Interior / boundary split	report interior error E^{int} ($i \le N-1$) and right-edge error E^{bnd} ($i = N$) separately — the direct evidence for "does the boundary closure limit the global error"
Mesh sequence	$N = 8, 16, 32, 64$ (four meshes; add 128 if time permits, to check the asymptotic range)
Order fitting	$p_k = \log_2\left(E_N / E_{2N}\right)$, the lecture formula $p_{\text{obs}} = \ln(E(h)/E(h/2))/\ln 2$; quick reference: error halved $\to p \approx 1$, quartered $\to p \approx 2$, eighthed $\to p \approx 3$; **identify the asymptotic range before fitting** (last 2–3 levels); coarse-mesh slopes are not evidence
Solver-accuracy gate	per mesh: iterative scaled residual $\le 10^{-10}$ (or direct solve), so that the algebraic error $\ll$ discretisation error (E2 is the exception: its tolerance is itself an object of study)

E1 PREDICTIONS | PREDICTIONS FIRST

① Closure A: boundary L_∞ order $\to 1$ (the explicit expectation of the pack's sanity check); global L_h^2 and L_∞ likewise expected to be order 1 — the first-order row acts like an $O(h)$ error layer at the right edge, decaying inward through the discrete Green function but not vanishing globally. **To be confirmed by measurement** — this is precisely what E1 answers (whether the boundary closure limits the global error).

② Closure B: with smooth data, global L_h^2 and L_∞ orders $\to 2$; boundary and interior errors of the same order.

③ The numerical solution satisfies the maximum principle: $u^*_{\min|\Gamma} \le U \le u^*_{\max|\Gamma}$ approximately holds (e^{x+y} monotone; minimum $e^0 = 1$, maximum near e^2).

4 · Algebraic Layer: Sparse Direct vs Iterative Solvers (Depth 2)

4.1 Sparse Direct Solver (SuperLU)

Primary choice `scipy.sparse.linalg.spsolve` (SuperLU, column permutation + fill-reducing ordering by default). Theoretical cost (2-D five-point grids):

Table 4.1 Direct-solver cost levels (main-problem size $n = N(N-1)$, bandwidth $b \approx N$)

Quantity	Banded view	Nested-dissection view (theoretical best ordering)
Fill-in	$O(n\,b) = O(N^3)$ stored entries	$O(N^2 \log N)$
Floating-point work	banded elimination $O(n\,b^2) = O(N^4)$ (ignores in-band sparsity)	fill-reducing ordering $O(N^3)$
Expected wall time	milliseconds to seconds for $N \le 512$ (C + BLAS, tiny constants)	

The two columns correspond to "the banded algorithm (in-band sparsity unused)" and "fill-reducing ordering (what SuperLU actually does)". Predictions are anchored on **measured wall time**; the theory column explains the **growth slope**.

No-tolerance reporting protocol (standing protocol): report the backward residual $\|b - A\hat{x}\|_2 / \|b\|_2$ (should reach $O(10^{-14})$), a condition-number estimate (`onenormest` cross-checked against the theoretical $\kappa \approx c\,h^{-2}$), and an independent cross-check of the forward error by one very-tight iterative solve.

4.2 Stationary Iterations: The Splitting Framework

Let $A = D - E - F$ (D diagonal, $-E$ strictly lower, $-F$ strictly upper); every scheme is $x^{(k+1)} = Gx^{(k)} + c$:

$$\text{Jacobi: } x^{(k+1)} = D^{-1}\left[b - (E+F)x^{(k)}\right] \qquad \text{GS: } (D-E)x^{(k+1)} = b + Fx^{(k)}$$

$$\text{SOR: } \left(\frac{D}{\omega} - E\right)x^{(k+1)} = \left[\left(\frac{1}{\omega} - 1\right)D + F\right]x^{(k)} + b, \qquad 0 < \omega < 2$$

Stopping criterion (lecture convention vs project stipulation): the lecture notes uniformly use the **iterate-difference test** $\|U^{(k+1)} - U^{(k)}\|_2 < \varepsilon$ ($\varepsilon = 10^{-6}$, compared after each full sweep against the saved old vector), with the note that "this criterion measures the update size, not the error against the exact solution". Project E2 follows the pack and uses the **relative residual** $\|b - Ax^{(k)}\|_2 / \|b\|_2 \le \text{tol}$ (pack wording: residual, not iterate difference) — because the present matrix is **nonsymmetric with widely differing row scalings** (h^{-2} PDE rows mixed with h^{-1}

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boundary rows), the ratio between iterate difference and true error is no longer controlled by the spectral radius alone, and a small iterate difference does not reliably bound the algebraic error. Execution: **primary criterion = scaled residual** (project convention); **comparison column = lecture iterate difference**. **Prediction**: on fine meshes with $\rho \to 1$ the iterate-difference test "false-converges" (stops early while the algebraic error is still large, since $\|\Delta U^{(k)}\| \sim (1-\rho)\,\|e^{(k)}\|$); the gap between the true scaled residuals at the two stopping points is itself a reportable E2 observation.

4.3 Spectral Radii and Iteration-Count Predictions

For the **pure-Dirichlet model problem** ($m \times m$ interior grid, $h = 1/(m+1)$) the classical closed forms hold (derivation sketch in Appendix B). Notation aligned with the lecture notes: the Jacobi iteration matrix is $J = I - D^{-1}A$ (identical to our G_J), $\rho(J) = \max_\ell |\lambda_\ell(J)|$ measuring the asymptotic error decay; the lecture ω_{opt} table ($h = \frac{1}{8} \to 1.4465$, $h = \frac{1}{16} \to 1.6735$, $h = \frac{1}{32} \to 1.8215$, $h = \frac{1}{64} \to 1.9065$, $h = \frac{1}{128} \to 1.9521$) matches our table below row by row under $h = 1/N$. The eigenvectors $\sin(ip\pi h)\sin(jq\pi h)$ separate variables and give

$$\rho_J = \cos(\pi h) \approx 1 - \frac{\pi^2 h^2}{2}, \qquad \rho_{GS} = \rho_J^2 = \cos^2(\pi h), \qquad \omega^* = \frac{2}{1+\sin(\pi h)}, \qquad \rho_{SOR}(\omega^*) = \omega^* - 1 \approx 1 - 2\pi h.$$

The number of iterations to reach a relative residual ε is $n_{\text{it}} \approx \ln(1/\varepsilon)\,/\,(-\ln\rho)$; the $\varepsilon = 10^{-8}$ prediction table:

Table 4.2 Iteration-count predictions (model-problem theory; calibrated by measurement on the mixed problem)

N	**Model size m^2 ($m = N-1$)**	**Jacobi ρ_J / steps**	**Gauss–Seidel steps**	**SOR(ω^*) ω^* / steps**	**Per-step cost**
16	225	0.98079 / ~950	~475	1.6736 / ~47	$\approx 5n$ flops (sparse matvec)
32	961	0.99518 / ~3 800	~1 900	1.8214 / ~94	idem
64	3 969	0.99879 / ~15 300	~7 600	1.9064 / ~188	
128	16 129	0.99970 / ~61 100	~30 600	1.9521 / ~375	
256	65 025	0.999925 / ~244 600	~122 300	1.9757 / ~747	idem

Steps computed with $n_{\text{it}} = \ln(1/\varepsilon)/(-\ln\rho)$, $\varepsilon = 10^{-8}$ ($\ln 10^8 \approx 18.42$); the size column is the interior-DOF count $(N-1)^2$ of the **pure-Dirichlet model problem**; E2 measures the main-problem matrix (size $N(N-1)$) and calibrates with small-mesh numerical spectral radii. Three slopes: Jacobi $O(h^{-2})$, GS $O(h^{-2})$ (halved constant), SOR $O(h^{-1})$ — **the predicted slopes of the E2 "iterations vs mesh" log-log plot**.

4.4 Adaptation to the Mixed-Boundary Matrix

- The matrix is **nonsymmetric**: the closed forms above no longer hold exactly, but the §3.6 properties (**M-matrix + irreducible weak dominance**) guarantee convergence of Jacobi and GS by Varga's theorem (spectral radius of the iteration matrix < 1); SOR converges for $0 < \omega \le 1$ (Ostrowski), while $\omega > 1$ must be measured.
- **ω protocol**: start from the model formula $\omega^*(h) = 2/(1+\sin\pi h)$ and scan $[\omega^* - 0.05,\ \omega^* + 0.03]$ on each mesh, taking the measured optimum (fewest steps); the scan curve goes into the report. In the pure-Neumann optional depth (before gauging), $\omega > 1$ must not be chosen freely — gauge first.
- **Numerical spectral radii**: on small meshes ($N \le 32$) compute the spectral radius of the iteration matrix $G = M^{-1}N$ by dense eigenvalues (`numpy.linalg.eigvals`) and compare with Table 4.2 — the bridge evidence from theory to the practical problem.
- **CG admissibility check (direct application of the lecture criterion)**: the lecture CG criterion is $A^\top = A$, $z^\top A z > 0$ ($z \ne 0$) (SPD). The main-problem matrix fails the first condition — the E1 measurement $\delta_A > 0$ is the quantitative evidence, and "the continuous Laplacian is self-adjoint" cannot replace "the discrete matrix is symmetric" (the error the pack names; the lecture criterion proves it). **The E3 benchmark** (rectangular pure Dirichlet; after elimination $A = K_{II}$ is SPD) **does satisfy the lecture CG condition** — E3's solver may add a CG column (`scipy.sparse.linalg.cg`), which together with E2 puts all four lecture iteration methods to work: Jacobi/GS/SOR on the nonsymmetric main problem, CG on the SPD benchmark; if CG is added, still stop on residuals per the pack and report iteration counts.
- **Krylov comparison (optional bonus)**: for the main problem the proper route is GMRES/BiCGSTAB + incomplete-LU preconditioning; a comparison column against the stationary methods explains "why the pack allows but does not require it".

4.5 E2: Matched-Accuracy Comparison Protocol

- **Matched accuracy target**: with "algebraic error within 1% of the discretisation error" as the standard, take $\text{tol} = 10^{-10}$ (relative residual) so all iterative solvers and the direct solve sit on one accuracy platform; add a $\text{tol} = 10^{-6}$ group to see whether a looser target changes the ranking.
- **Reported quantities**: wall time (`time.perf_counter`), iteration counts, (direct) fill-in nonzeros; log-log plots of time vs N and steps vs N.
- **Crossover**: report the N at which "iterative time = direct time", if it exists; **if no crossover occurs over the whole tested range, report that explicitly** (pack requirement).
- **Fairness controls**: same assembled matrix, same initial guess $x^{(0)} = 0$ (or one declared warm-start rule), same stopping-code path; iterative implementations must be **vectorised** (red-black Jacobi/GS, vectorised SOR) or explicitly flagged as pure loops with their language overhead reported — otherwise the wall-time comparison is meaningless.

> **E2 PREDICTIONS | PREDICTIONS FIRST**
>
> ① Ranking (wall time): direct < SOR < GS < Jacobi; the direct solver is expected to win over the **entire tested range** $N \le 256$ — the theory slopes (direct $O(N^3)$, SOR total $O(N^3 \ln)$, Jacobi $O(N^4 \ln)$) and the implementation constants (C/BLAS vs Python loops) both point that way; **no crossover is expected and will be reported explicitly**.
>
> ② Step-count slopes: Jacobi/GS ≈ -2, SOR ≈ -1 (slopes of $\log n_{\text{it}}$ vs $\log N$).
>
> ③ **Theoretical existence of a crossover**: direct $O(N^3)$ vs SOR $O(N^3 \ln)$ are of the same order with a constant advantage for the direct method $\Rightarrow$ any asymptotic crossover lies far beyond $N \gg 256$; memory (fill $O(N^2 \log N)$) becomes the direct method's bottleneck first — a forward-looking remark for the cost-vs-accuracy discussion.

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5 · FDM vs FEM on a Common Benchmark (Depth 3)

5.1 Why a Separate Benchmark Is Prescribed

Direction 2 (like the disk direction) **prescribes the FDM boundary closures** — the FEM treatment of mixed data (Neumann/Robin as natural boundary conditions in the weak form) is a different methodology, and comparing it directly against the prescribed FDM closures would confound "discretisation differences" with "boundary-treatment differences". The pack therefore stipulates: the FDM vs FEM comparison runs on **an independent rectangular pure-Dirichlet Poisson benchmark** in which neither method is subject to special constraints — this is the origin of the "build a separate rectangular Poisson benchmark" clause in the task list.

5.2 Benchmark Specification

$$-\Delta u = f \text{ in } \Omega = (0,1)^2, \qquad u = g \text{ on } \partial\Omega, \qquad u^* = e^{x+y},\ \ f = -2e^{x+y},\ \ g = u^*|_{\partial\Omega}.$$

We reuse the main problem's manufactured solution (unified narrative; the right-hand-side and error code carry over directly); during implementation a second data point $u^* = \sin \pi x \sin \pi y$ ($g \equiv 0$) provides a cross-check. Data are verified against `03_Scaffolding/test_problem_bank.py` before use.

5.3 Two Discretisations and the Coincident-Node Device

- **FDM side**: five-point stencil on the $N \times N$ grid; interior unknowns $(N-1)^2$.
- **FEM side**: continuous piecewise-linear P1 elements; **cut each of the same $N \times N$ squares along the same diagonal into two triangles**, producing a shape-regular, consistent triangulation. Nodes = all $(N+1)^2$ grid points; after strong imposition (elimination) of the Dirichlet data the interior DOFs are exactly $(N-1)^2$ — **the same DOF count as FDM on a fully coincident node set**.

> **COINCIDENT-NODE DEVICE**
>
> Coincident nodes give three benefits at once: ① the common-node L_∞ comparison needs no projection or interpolation; ② identical h and DOFs make the comparison table automatically fair; ③ total DOFs drop out as a confounding variable. The pack's "matched mesh scale, same L2 norm, comparable tolerances" is satisfied in one stroke.

FEM modelling points (the FEM counterpart of Part 2, to be finalised after Lecture 4; notation follows the FEM lecture notes: local shape functions $N_a^{(e)}$, global basis φ_i, nodal approximation $u_h = \sum_j U_j \varphi_j$):

- **Why the weak form is unavoidable** (lecture narrative, for the report background): piecewise-linear u_h satisfies $u_h'' = 0$ inside each 1-D cell (or $\Delta u_h = 0$ inside each triangle) and **cannot satisfy the second-order strong form pointwise**; integration by parts transfers one derivative to the test function (the §1.2 chain), after which only piecewise-constant first derivatives are needed — which is exactly why FEM "naturally" accommodates Neumann/Robin data while FDM needs a dedicated closure;
- **Galerkin discretisation** (lecture notation): take $v = \varphi_i$ for every free node and insert the nodal approximation, giving the stiffness and load $K_{ij} = \int_\Omega \nabla\varphi_i \cdot \nabla\varphi_j \, dx$, $F_i = \int_\Omega f\varphi_i \, dx$;
- **Dirichlet elimination** (lecture formula): with free node set I and known boundary node set B, assembly gives $K_{II}U_I + K_{IB}\,g_B = F_I$, i.e. the lecture form $\underbrace{K_{II}}_{A} U_I = \underbrace{F_I - K_{IB}\,g_B}_{b}$ — which connects directly to the pack's $Ax = b$; if $g \equiv 0$ then $b = F_I$. The boundary data are imposed by interpolation $u_h|_{\partial\Omega} = I_h g$;
- **Element stiffness and the exact matrix identity** (right-isosceles triangle, legs h, area $h^2/2$; local node 1 at the **right angle**): Unknown environment 'psmallmatrix' . On the fixed-diagonal triangulation, in every element the two **hypotenuse-end basis gradients are orthogonal** (e.g. $(1/h, 0) \cdot (0, 1/h) = 0$), so all diagonal-neighbour couplings vanish, and the global interior row is exactly the five-point pattern scaled by h^2:

  $$h^2\, A_{\mathrm{FDM}} \;=\; K_{P1},$$

  i.e. identical sparsity patterns (row density 5) — verified numerically by the team's assembly audit (relative discrepancy 2.0×10^{-15}, report Eq (20)). **Correction note**: an earlier modelling draft had predicted extra -1 diagonal couplings (row density 7); the audit falsified this — see the ICS log, item A3. The residual FDM–P1 difference lies entirely in the **load vectors** (b_{P1} carries integrated moments $\int f\varphi_i \, dx$ while b_{FDM} carries nodal samples f_i), which explains the measured $\approx 2\times$ error-constant gap for the exponential data;
- implementation choice **scikit-fem** (the pack: "the comparison itself requires a 2D implementation such as scikit-fem"; `fem1d_skeleton.py` may be consulted for assembly ideas only — a 1-D FEM computation does not constitute the comparison);
- load integration uses a sufficiently accurate quadrature ($u^* = e^{x+y}$ is smooth; three or more points keep the load error $\ll O(h^2)$).

5.4 Matched Protocol and Predictions

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Table 5.1 E3 comparison protocol (itemised implementation of the pack's requirements)

Controlled variable	Protocol
Mesh scale	same $h = 1/N$; mesh family $N = 8, 16, 32, 64$
DOFs	$(N-1)^2$ on both sides (reported separately against implementation drift)
Norms	common-node error vectors: L^2_h (h^2 weights) and L_∞; FEM additionally reports the integral L^2 (mass-matrix definition) as a supplement
Algebraic tolerance	direct solver (spsolve) on both sides; backward residuals reported for each
Reported quantities	assembly time, solve time, nonzeros, L^2_h/L_∞ errors, observed orders — **reported separately**, never merged into a single "total time"
Attribution discipline	a difference may be attributed to "the method" only when data, dimension, mesh scale, norm and stopping criterion all match (the operative version of the pack's wording)

E3 PREDICTIONS | PREDICTIONS FIRST

① Both L^2_h error orders $\to 2$ (convex domain, smooth solution; FEM theory $O(h^2)$ plus nodal superconvergence); L_∞ of the same order.

② Error constants: with the exponential data the measured P1 constant is $\approx 2\times$ the FDM constant — **attributed to the load vectors** (integrated moments vs nodal samples), since $h^2 A_{\text{FDM}} = K_{P1}$; with the sine data the two constants nearly coincide. Measured outcome decides.

③ Nonzeros: identical patterns, $\approx 5(N-1)^2$ on both sides; assembly time longer for FEM (element-wise integration); solve times on the same matrix are trivially comparable (report this explicitly, together with the separate assembly timings, as the pack's "report assembly time, solve time, nonzeros ... separately" requires).

6 · Regularity and Order Degradation (Depth 4)

6.1 How Regularity Sets the Rate

The error bound $u - u_h$ has two components: **Céa's lemma (stability)** + **interpolation error (approximability)**. Approximability is governed by the Sobolev regularity of u: for linear elements (FEM) and second-order schemes (FDM), a smooth solution $u \in H^2$ gives energy norm $O(h)$ and L^2 norm $O(h^2)$ (the latter via Aubin–Nitsche duality). If $u \notin H^2$, say $u \in H^{1+s}$ ($0 < s \leq 1$):

$$\|\nabla(u-u_h)\|_{L^2} = O(h^s), \qquad \|u-u_h\|_{L^2} = O\big(h^{\min(2s,\,s+1)}\big) \ \Big(= O(h^{2s}) \text{ when } s<1\Big),$$

where the L^2 rate uses the fact that **the dual problem has the same limited regularity** — on re-entrant-corner domains the dual solution is also only in H^{1+s}, so Aubin–Nitsche earns only one more factor h^s. On the L-shaped domain this mechanism produces the famous "degraded pair": $s = 2/3 \Rightarrow$ energy $O(h^{2/3})$, L^2 $O(h^{4/3})$, instead of the smooth-case $O(h)$ and $O(h^2)$. **Depth 4 consists precisely of measuring this pair and explaining the mechanism.**

6.2 Design: Same Solver, Same Domain, Two Solutions

Key design decision: place the smooth-vs-nonsmooth comparison **on the L-shaped domain** (rather than manufacturing an artificial singular solution on the square), using **two analytic solutions, both with** $f = 0$:

Table 6.1 The E4 pair (literally "same solver, one smooth + one nonsmooth")

Solution	Expression	Regularity	Predicted orders (uniform meshes)
Smooth control	$u_s = e^x \cos y$ (harmonic, $f=0$)	$C^\infty(\bar\Omega)$	$L_\infty, L^2_h \to 2$ (no degradation)
Singular solution	$u^* = r^{2/3}\sin\frac{2\varphi}{3}$ (harmonic, $f=0$)	$H^{1+s}\ \forall s < \frac{2}{3}; \notin H^2$	$L_\infty \to \frac{2}{3}$ (diagnostic), $L^2_h \to \frac{4}{3}$ (measured FDM)

Both right-hand sides vanish, both are pure Dirichlet data — **the only variable is the regularity of the solution**, leaving no confounding room in attributing the degradation. This is the architecture in which Depth 4 and the innovation share one code line: E4 runs both solutions on uniform meshes; E5 runs graded meshes on the same singular solution.

7 · Innovation: L-Shaped Domain, Corner Singularity, and Graded Meshes

7.1 Corner-Singularity Theory: Why Exactly $2/3$

The classical elliptic-regularity result (Kondratiev): on **convex** polygonal domains, solutions of $-\Delta u = f \in L^2$ belong to H^2, and second-order schemes attain their full order; at a **re-entrant corner** the solution develops a power-law singularity. Separation of variables in polar coordinates (a theoretical highlight for the report): among the harmonic family $r^\lambda \sin(\lambda\varphi)$, vanishing on both boundary rays $\varphi = 0$ and $\varphi = \omega$ requires $\lambda\omega = k\pi$, and the smallest exponent is

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$$\lambda_1 = \frac{\pi}{\omega}; \qquad \text{L-shaped domain } \Omega = (-1,1)^2 \setminus [0,1)^2 : \ \omega = \frac{3\pi}{2} \implies \lambda_1 = \frac{2}{3}, \qquad u^*(r,\varphi) = r^{2/3}\sin\frac{2\varphi}{3}.$$

Verification points: u^* is harmonic away from the corner ($\Delta r^\lambda \sin\lambda\varphi \equiv 0$ for $r>0$); $u^*=0$ on both interior rays; $u^* \in H^{1+s}(\Omega)$ for every $s<2/3$ ($\nabla u^* \sim r^{-1/3}$ integrable, $D^2u^* \sim r^{-4/3}$ not), but $u^* \notin H^2(\Omega)$. **The regularity exponent hooks directly into the convergence-order exponent** ($s=\lambda_1$ in §6.1) — that is the whole mathematics of the "degradation phenomenon".

7.2 Implementation Spec and Pitfalls

- **Branch convention** (the pack requires it to be implemented explicitly): $\theta = \operatorname{atan2}(y,x)$, add 2π whenever $\theta < \pi/2$, so $\theta \in [\pi/2, 2\pi]$ and $\varphi = \theta - \pi/2 \in [0, 3\pi/2]$. This convention makes $u^*=0$ on both interior rays (the segment $y=0, x\in(0,1)$ and the segment $x=0, y\in(0,1)$) — **a wrong branch shows up immediately in the error plot** (a spurious nonzero near the corner), which is itself a built-in sanity check.
- **Data**: $f=0$; all boundary data are analytic samples: $g=0$ on the interior rays; $g = u^*|_{\partial\Omega}$ on the outer boundary (the segments $x=-1, y=-1, y=1$ for $x\in[-1,0]$, and $x=1$ for $y\in[-1,0]$). **Independent verifications** for the report: assembly consistency of $f=0$ (the RHS carries only boundary data), and pointwise checks of the two zero-trace rays.
- **Prohibition** (pack wording): **never evaluate the singular gradient at the corner** — any gradient/energy post-processing is undefined at $(0,0)$; even the discrete energy-norm sums must be annotated near the corner ($\nabla u^* \sim h^{-1/3}$ at $r \geq h$: finite but large).
- **Grid alignment**: $x_i = -1 + ih$, $h = 2/N$ with N **even** (32, 64, 128, 256), so $x=0$ and $y=0$ are grid lines; the removed block $[0,1)\times[0,1)$ consists of nodes with $i \geq N/2$ and $j \geq N/2$ (Dirichlet nodes on the block boundary excluded). Unknowns are the interior nodes of the L-shape. Five-point rows are assembled only at nodes whose four neighbours are either interior or Dirichlet — the re-entrant corner creates no irregular rows on an aligned grid (a structural advantage of FDM worth stating in the report).

7.3 E4: Uniform-Mesh Convergence Test (FDM)

> **E4 PREDICTIONS | PREDICTIONS FIRST**
>
> ① Smooth control $e^x\cos y$: L_∞ and L_h^2 orders $\to 2$ (confirms the solver is healthy).
>
> ② Singular $r^{2/3}\sin\frac{2\varphi}{3}$: L_∞ order $\to \frac{2}{3}$ (corner neighbourhood dominates; the max error sits at distance $O(h)$ from the corner, of size $\sim h^{2/3}$, possibly with a log correction); L_h^2 order $\to \frac{4}{3}$. The degradation relative to order 2 is **the core Depth-4 conclusion**, cross-checked against the FEM theory rates ($\frac{2}{3}$ energy / $\frac{4}{3}$ L^2) — but the report must label these as **measured FDM results** (FDM has no matching a priori theory; the FEM argument cannot be quoted directly).
>
> ③ Spatial distribution: the error map shows the error hugging the re-entrant corner and the peak shrinking towards the corner under refinement — the degradation is **a local phenomenon**, which provides the geometric motivation for graded meshes.

7.4 E5: Graded Meshes — Threshold and Implementation Routes

Why grading restores the optimal rate (the scaling argument, a derivation highlight for the report): let the element size near the corner satisfy $h_T(r) \asymp h\, r^{1-1/\mu}$ (radial grading exponent $\mu \geq 1$, $\mu=1$ uniform; r a dimensionless local radius). The per-element interpolation error obeys $\|u^* - I_h u^*\|^2_{H^1(T)} \lesssim h_T^4\, r^{2\lambda-4}$; summing with the element density $h_T^{-2}\, r\, dr$ gives

$$\|\nabla(u^*-u_h)\|_{L^2}^2 \lesssim h^2 \int_0^\varepsilon r^{\,2\lambda-1-2/\mu}\, dr \implies \begin{cases} \mu > 1/\lambda: & \|e\|_{H^1} = O(h) \qquad \text{(integral converges; optimal rate restored)} \\ \mu = 1: & \|e\|_{H^1}^2 \sim h^2 \cdot h^{2\lambda-2} \ \Rightarrow\ \|e\|_{H^1} \sim h^\lambda \quad \text{(uniform-mesh degraded rate)} \end{cases}$$

Threshold $\mu^* = 1/\lambda = \frac{3}{2}$: for $\mu > \mu^*$ the corner contribution stays bounded and the energy error is governed by the "smooth region" at rate $O(h)$; substituting $\mu=1$ reproduces the uniform-grid $h^{2/3}$ exactly, so the argument is self-consistent. In the DOF-error currency prescribed by the pack:

$$\text{uniform (quasi-uniform P1)}: \quad \|\nabla e\|_{L^2} = O\big(N_{\text{dof}}^{-1/3}\big) \xrightarrow{\text{grading } \mu>\mu^*} O\big(N_{\text{dof}}^{-1/2}\big) \text{ (optimal)}$$

Table 7.1 Two implementation routes (A recommended; B validates the theory)

Route	Construction	Advantages / risks
A. Red-refinement pyramid (recommended)	start from the uniform L-shaped triangulation; repeatedly quarter all triangles with distance-to-corner $< \delta$ (right-isosceles red refinement is shape-preserving), then run a **closure loop to remove hanging nodes**; sweep the levels $\kappa = 2, 3, 4$	shape-regularity is preserved strictly; scikit-fem assembles directly (coordinates + triangle table built by hand); size law $h(r) \sim (r/\delta)\, h_0$, i.e. grading strength **exceeds** the theoretical need $h^*(r) \propto r^{1-1/\mu^*} = r^{1/3}$ (exponent 1 > 1/3: optimal rate restored, slightly more DOFs). Risk: closure logic needs care (leftover hanging nodes degrade the P1 assembly)
B. Radially graded polar mesh (validates the μ theory)	$r_i = (i/n)^\mu$ with equispaced angles → Cartesian coordinates → ring-wise triangulation; sweep $\mu \in \{1, 1.25, 1.5, 2, 3\}$	directly confronts the $\mu^* = 3/2$ threshold; one "μ-recovery curve" is the theory-validation plot. Risk: elements near the corner are anisotropic ($h_\varphi/h_r \to 0$); the loss of shape regularity must be reported honestly (Apel's anisotropic theory supports effectiveness, but the currency differs from route A)
C. Adaptive mark–refine (stretch goal)	residual indicator $\eta_T^2 = \frac{1}{2}\sum_{E\subset\partial T} h_E \|[\![\nabla u_h \cdot t]\!]\|_E^2$ (pure jump term since $f=0$); Dörfler marking $\theta = 0.2$–0.5; red refinement	absorbs open direction 2 ("adaptive vs uniform"): adaptivity wins on singular problems and wastes work on smooth ones — quantify with one extra $e^x\cos y$ control. Risk: needs a mesh library or a hand-written refinement manager (drop if outside the time box)

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Acceptance criterion (E5 plotting protocol): on the same singular solution, the same norm definitions and the same solver tolerance, plot $\|e\|$ vs N_{dof} on log-log axes (uniform / graded A / best-μ graded B), with **measured slopes and theory lines shown side by side**:

$$\text{energy norm: uniform } N_{\mathrm{dof}}^{-1/3} \;\to\; \text{graded optimal } N_{\mathrm{dof}}^{-1/2}; \qquad L^2 \text{ norm: uniform } N_{\mathrm{dof}}^{-2/3} \;\to\; \text{graded optimal } N_{\mathrm{dof}}^{-1}.$$

The L^2 currency follows from $h^{4/3}$ and $N_{\mathrm{dof}} \sim h^{-2}$ (Aubin–Nitsche). Any deviation of the measured slope from the theory line must be explained — exactly the pack's "measure rather than assume" attitude.

Quadrature warning (pack requirement): near the corner, load and error integrals must use quadrature matched to the local element size (scikit-fem's default low order suffices for the P1 load of smooth data, but the **error integrals** $\int (u^* - u_h)^2$ face an integrand $\sim r^{4/3}$ — integrable — while the squared gradient $\sim r^{-2/3}$ is **not**: energy-error integration must exclude the corner element or use adaptive quadrature, with the treatment stated in the report).

8 · Experiment Matrix and Master Prediction Table

Table 8.1 Full experiment matrix (E1–E6; outputs = the minimal report/code artefacts)

ID	Name	Meshes / parameters	Outputs	Predicted outcome (acceptance line)
E1	Two closures × two variants (core)	$N = 8, 16, 32, 64$; {Neumann, Robin}×{A, B}	4 error tables (L_h^2, L_∞, interior/edge split) + fitted orders; spy plot; δ_A table; scaled-residual table; u^* substitution audit (Appendix A cards)	A: boundary $L_\infty \to 1$; B: global $\to 2$; maximum principle holds
E2	Direct vs Jacobi/GS/SOR	$N = 16..256$; tol $= 10^{-10}$ (+ a 10^{-6} group); ω scan	time/steps vs N log-log plots; crossover determination; small-mesh spectral radii vs theory	step slopes $-2/-2/-1$; direct wins throughout, no crossover (report explicitly)
E3	FDM vs FEM (scikit-fem)	$N = 8..64$; $u^* = e^{x+y} + \sin \pi x \sin \pi y$	assembly/solve time, nnz, errors, orders — FDM/FEM in separate columns; common-node L_∞; matrix-pattern audit	both $L_h^2 \to 2$; identical patterns ($h^2 A_{\mathrm{FDM}} = K_{P1}$); constant gap from load vectors
E4	Smooth vs singular (L-domain, FDM)	$h = 2/N$, $N = 32..256$ (even)	two-solution convergence tables + error maps (corner localisation)	smooth→2; singular $L_\infty \to 2/3$, $L_h^2 \to 4/3$
E5	Graded meshes (L-domain, FEM)	red refinement $\kappa = 2, 3, 4$; radial μ sweep; adaptive stretch	$\|e\|$ vs N_{dof} three-curve log-log plot with measured/theory slopes; mesh plots; indicator values	uniform $N_{\mathrm{dof}}^{-1/3} \to$ graded recovers $N_{\mathrm{dof}}^{-1/2}$ (energy); validate $\mu^* = 3/2$
E6	Pure-Neumann solvability (optional depth)	turn Γ_D into Neumann; eigenvalue check	compatibility-defect quantification; demonstration of $A\mathbf{1} \approx \mathbf{0}$ before the gauge; pre/post-gauge comparison	defect converges to a small quantity; gauge restores uniqueness

Interface between roles: E1–E2 belong to the main-problem code line (FDM assembly + solvers), E3–E5 to the benchmark/innovation line (scikit-fem), and E4 reuses the main-problem FDM assembly (new domain + new data functions). The two lines are developed in parallel over shared modules: index mapping, error metrics, order fitting, log-log plotting.

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9 · Risk and Pitfall Checklist

#	Pitfall	Countermeasure (into the code checklist)
1	Treating the algebraic residual as the discretisation error	define and report both separately (§3.7 scaled residual + §3.8 error); tighten-tolerance re-solves confirm the plateau
2	A corner overwritten by a Neumann/Robin row	the unknown set structurally excludes right-edge nodes with $j = 0, N$ (§3.3 assembly rule)
3	Using CG on a nonsymmetric matrix	primary solver spsolve; if Krylov is wanted use GMRES/BiCGSTAB, justified in the report by $\delta_A > 0$; CG only on the SPD E3 benchmark
4	Comparing unscaled residuals across meshes	mixed h^{-1} and h^{-2} rows: always scale by D_s first (§3.7)
5	Attributing every closure difference to "order 1 vs order 2"	state the constitution difference "derivative row replaces PDE row vs PDE row retained with ghost elimination" (§3.6)
6	Choosing a low-degree-polynomial test solution	fixed choices e^{x+y} / $\sin \pi x \sin \pi y$ / $e^x \cos y$ (none is exactly reproduced by the stencil)
7	Wrong branch angle on the L-domain	$\operatorname{atan2} + [\theta < \pi/2] \cdot 2\pi$; unit test asserts $u^* \equiv 0$ pointwise on the rays
8	Evaluating the singular gradient at the corner	post-processing explicitly excludes $(0, 0)$; energy integrals avoid the corner or declare the treatment (§7.4 quadrature warning)
9	Pure Neumann (if E6 is run) without the compatibility correction	quantify the discrete defect, project-correct the RHS, gauge, then solve
10	Iterative solvers as pure Python loops	vectorise (red-black ordering) or report the language overhead explicitly, to keep E2 wall times fair
11	Using the SOR ω formula without measurement	theoretical initial value + small scan; the scan curve goes into the report
12	Leftover hanging nodes after red refinement	run the closure loop until none remain, then assemble
13	Order fitting on coarse-mesh slopes	fit only the asymptotic range (last 2–3 levels); display all levels
14	Reusing the lecture iterate-difference stopping rule in E2	E2's primary criterion is the scaled residual (pack stipulation); the iterate-difference column is reported as a comparison, together with the true residual at both stopping points (§4.2 "false convergence")

10 · Protocol Mapping and Implementation Interface

10.1 Mapping to Course Deliverables

Course deliverable	Where in this document	Timing
`problem_definition_sheet.tex` Part 1 (continuous problem, sign conventions, method choice)	all of §2 + §1.2	Week-5 issue ticket (with Part 2, FDM teams)
Part 2 (discrete operator: stencil)	all of §3 (including the boundary-row cards)	with Part 1
Pre-computation protocol (predictions before solves)	the **Predictions First** boxes of §3.8 / §4.5 / §7.3 / §7.4; transcribe into the worksheet before each new solve, **never edited afterwards**	before every new solve
AI five-gate audit (AI-generated solver code must pass)	performed during implementation: rows vs Appendix-A cards, u^* substitution, maximum principle, residual separation, prediction check	before code integration
Lecture notes (1-D differences; FEM & iterative solvers)	alignment points: bulk-difference/block-elimination equivalence (§3.4); ghost operation order and second-order origin (§3.5); weak-form chain and $K_{II}U_I = F_I - K_{IB}g_B$ (§1.2/§5.3); $J = I - D^{-1}A$ and the ω_{opt} table (§4.3/App. B); iterate-difference vs residual stopping (§4.2); CG admissibility check (§4.4); pure-Neumann shift argument (§2.3); p_{obs} formula (§3.8)	throughout the presentation and report

10.2 Implementation Interface (for the programming module)

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```
# shared modules
index_map(N)              -> (k_of[i,j], node_type[i,j])      # unknown/Dirichlet/corner ownership
assemble_core(N, bc_variant, closure, data) -> (A_csr, b)    # E1 four assemblies: {neu,rob} x {onesided, ghost}
exact_u(x, y) / f / g / q / r                                # main-problem MMS data (one set of functions)
error_metrics(U, u_star, mask) -> (L2h, Linf, E_int, E_bnd)  # §3.8 conventions
fit_order(Ns, Es) -> (orders, asymptotic_slope)              # asymptotic-range fitting

# E2
solve_direct(A, b) / jacobi(A, b, tol) / gauss_seidel(...) / sor(A, b, omega, tol)
spectral_radius(G_matrix, N_small)                           # small-mesh eigenvalue measurement
# E3 (scikit-fem)
fem_poisson_2d(N, u_star) -> (u_h_nodes, K_nnz, t_assemble, t_solve)
# E4/E5
Lshape_grid(N) / Lshape_exact_s(x,y) / Lshape_exact_u(x,y)  # branch convention built in + unit tests
red_refine(mesh, kappa, delta) -> mesh_closed               # red refinement + closure
radial_graded_mesh(mu, n) -> mesh                            # E5-B
refinement_indicator(u_h, mesh) -> eta_T                     # E5-C (stretch)
# output: one CSV per experiment (columns: N, n_dof, nnz, L2h, Linf, E_int, E_bnd, resid_scaled, t_solve, iters, order)
```

Dependencies: `numpy / scipy` (main problem, E2), `scikit-fem` (E3/E5, needs installing), `matplotlib` (plots). Vectorise E2's iterations (red-black); pass every experiment through the five-gate audit on a small case before the four-mesh runs.

Appendix A · Boundary Row Cards

The four right-edge rows ($1 \le j \le N-1$; k the index of $U_{N,j}$, k_W that of $U_{N-1,j}$), for line-by-line assembly and auditing; the u^* substitution column gives the expected "hand-computed" residual.

Table A.1 Interior row and four boundary rows (matrix form; scaled by h^2 where noted)

Row	Matrix row (LHS)	RHS	u^* substitution audit (local residual)
Interior PDE row (incl. $i = N-1$)	$\frac{1}{h^2}(4U_P - U_W - U_E - U_S - U_N)$ at $i = N-1$ the eastern entry $U_E = U_{N,j}$ is a right-edge **unknown**	$f_{i,j} + \frac{1}{h^2}\sum$(known Dirichlet neighbours)	$-\frac{h^2}{6}e^{x_i+y_j} + O(h^4)$
A · Neumann one-sided	$\frac{1}{h}U_{N,j} - \frac{1}{h}U_{N-1,j}$	$q_j = e^{1+y_j}$	$-e^{1+y_j}\frac{h}{2} + O(h^2) < 0$ (first order)
A · Robin one-sided	$(\frac{1}{h}+2)U_{N,j} - \frac{1}{h}U_{N-1,j}$	$r_j = 3e^{1+y_j}$	$\frac{1}{h}e^{1+y}\big[(1-e^{-h})+2h\big] - 3e^{1+y} = -e^{1+y_j}\frac{h}{2}(1+O(h))$ (first order)
B · Neumann ghost ($\times h^2$)	$4U_{N,j} - 2U_{N-1,j} - U_{N,j-1} - U_{N,j+1}$	$h^2 f_{N,j} + 2h\,q_j$	BC component $e^{1+y_j}\frac{h^2}{6}$; PDE component $-\frac{h^2}{6}e^{1+y_j}$; **complete row $O(h)$: $(2/h)$-amplified sum** $\approx e^{1+y_j}(\frac{h}{3} - \frac{h^2}{6})$
B · Robin ghost ($\times h^2$)	$(4+2h\beta)U_{N,j} - 2U_{N-1,j} - U_{N,j-1} - U_{N,j+1}$ $\beta = 2 \Rightarrow$ diagonal $4+4h$, row sum $4h > 0$ strictly dominant	$h^2 f_{N,j} + 2h\,r_j$	same structure; complete row $O(h)$ as above
Lecture bulk-substitution row reference only (not this project's convention)	$3U_{N-1,j} - U_{N-2,j} - U_{N-1,j-1} - U_{N-1,j+1}$ ($\times\frac{1}{h^2}$) the boundary node carries no unknown; $U_{N,j} = U_{N-1,j} + hq_j$ recovers it afterwards	$f_{N-1,j} + q_j/h$	block-elimination equivalent of closure A — same solution; symmetric matrix with $N-1$ fewer unknowns; if implemented by mistake, E1's right-edge error split and the $\|A - A^\top\|/\|A\|$ metric lose their meaning — return to the prescribed convention

Robin one-sided row, expanded: $\frac{u^*(1)-u^*(1-h)}{h} + \beta u^*(1) - r = \frac{e^{1+y}(1-e^{-h})}{h} + 2e^{1+y} - 3e^{1+y} = e^{1+y}\big[\frac{1-e^{-h}}{h} - 1\big] \approx -e^{1+y_j}\frac{h}{2}$ √ same leading term as the Neumann row (the Robin data are designed as $q + \beta u^*$, so the β contribution cancels exactly).

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Appendix B · Spectral-Radius Derivation Sketch

Pure-Dirichlet model problem, $m = N - 1$ interior nodes per direction. The sine basis $\phi_{pq}(i,j) = \sin(ip\pi h)\sin(jq\pi h)$ is a common eigenbasis of the discrete Laplacian and of the Jacobi iteration matrix (separability). Acting on ϕ_{pq}, the Jacobi iteration matrix $J = I - D^{-1}A$ returns at every node the eigenvalue

$$\mu_{pq} = \frac{1}{2}\big[\cos(p\pi h) + \cos(q\pi h)\big], \qquad p,q = 1,\dots,m \implies \rho_J = \mu_{11} = \cos(\pi h).$$

GS: for consistently ordered matrices the GS eigenvalues satisfy $\mu^{GS} = (\mu^J)^2$ (Young), hence $\rho_{GS} = \rho_J^2 = \cos^2(\pi h)$. **SOR**: the eigenvalue relation is the quadratic $(\mu^{SOR} + \omega - 1)^2 = \mu^{SOR}\omega^2(\mu^J)^2$; minimising the spectral radius gives $\omega^* = \frac{2}{1+\sqrt{1-\rho_J^2}} = \frac{2}{1+\sin\pi h}$ with $\rho_{SOR} = \omega^* - 1$ (a double root). Asymptotics: $\rho_J = 1 - \frac{\pi^2 h^2}{2} + O(h^4)$, $\rho_{GS} = 1 - \pi^2 h^2 + O(h^4)$, $\rho_{SOR} = 1 - 2\pi h + O(h^2)$.

The present matrices are nonsymmetric ⇒ the closed forms serve as theoretical anchors on the pure-Dirichlet subproblem only; on the mixed matrices they are calibrated by the numerical spectral radii of §4.4.

Appendix C · Master Prediction Cheat-Sheet

Table C.1 All falsifiable predictions at a glance (the drafting sheet for the pre-computation protocol)

Exp.	Prediction	Basis
E1	closure A: boundary L_∞ order $\to 1$; closure B: global $L_h^2/L_\infty \to 2$; A global order 1 (to be measured)	$O(h)$ vs $O(h^2)$ local consistency + discrete maximum principle
E1	δ_A (relative Frobenius nonsymmetry) decays with N but stays > 0	$O(N)$ mismatched boundary rows out of $O(N^2)$
E1	$\kappa(A) \sim c\,h^{-2}$	standard 2-D elliptic discretisation result
E2	step-count slopes: Jacobi/GS ≈ 2, SOR ≈ 1 (against $\ln N$)	Table 4.2 closed forms
E2	wall-time ranking spsolve < SOR < GS < Jacobi; no crossover for $N \le 256$	flop slopes + implementation constants
E2	iterate-difference stopping false-converges on fine meshes (stops early with a larger true residual)	$\|\Delta U^{(k)}\| \sim (1-\rho)\|e^{(k)}\|$ with $\rho \to 1$
E3	L_h^2 order $\to 2$ both; identical nonzero patterns ($h^2 A_{\mathrm{FDM}} = K_{P1}$); $\approx 2\times$ constant gap for exponential data from the load vectors	convex-domain smooth data + coincident-DOF design + assembly audit
E4	smooth order 2; singular $L_\infty \to 2/3$, $L_h^2 \to 4/3$ (labelled measured FDM)	regularity $\lambda_1 = \pi/\omega = 2/3$
E5	energy error: uniform $N_{\mathrm{dof}}^{-1/3}$ → graded $N_{\mathrm{dof}}^{-1/2}$; $\mu^* = 3/2$	§7.4 scaling argument
E6	$A\mathbf{1} \approx \mathbf{0}$ (pure-Neumann null vector); small compatibility defect	zero row sums + quadrature mismatch

End of the modelling document · Modelling & Programming module · Direction 2 (Mixed Boundary Data) · Next step: implementation (§10.2 interface) and the five-gate audit

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